Lab4-2 Caravel FIR

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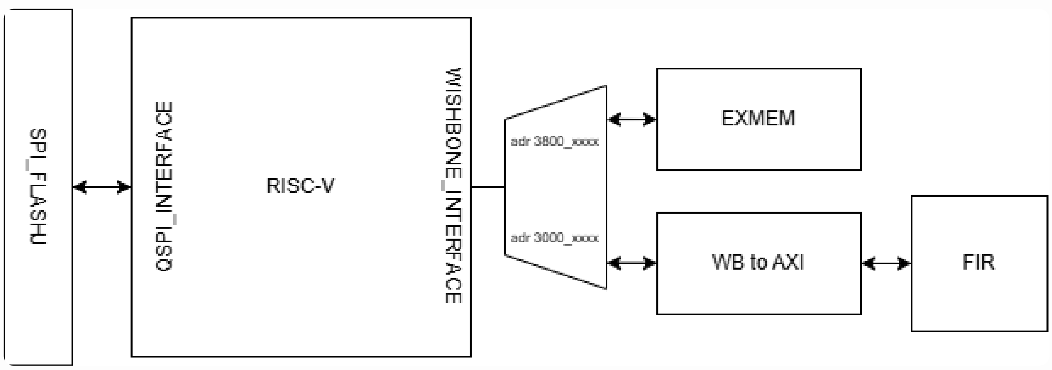
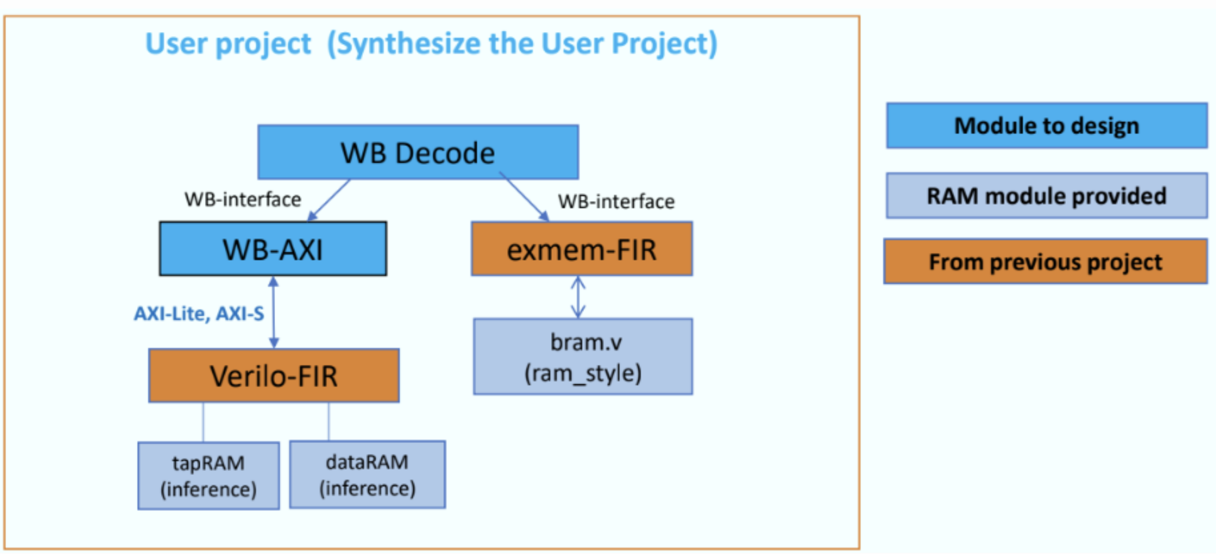
• Integrate Lab3-FIR & exmem-FIR (Lab4-1) into Caravel user project area (add WB interface)
• Execute RISC-V firmware (FIR) from user project memory
• Firmware to move data in/out FIR

Design block diagram – datapath, control-path



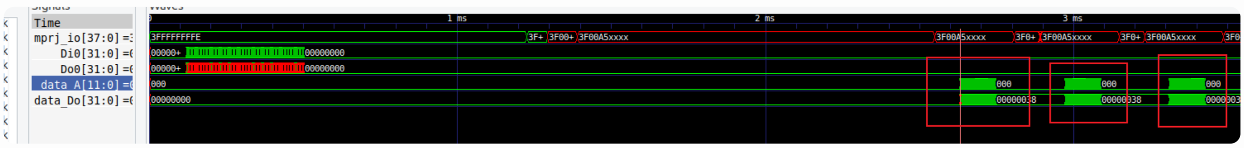
The interface protocol between firmware, user project, and testbench



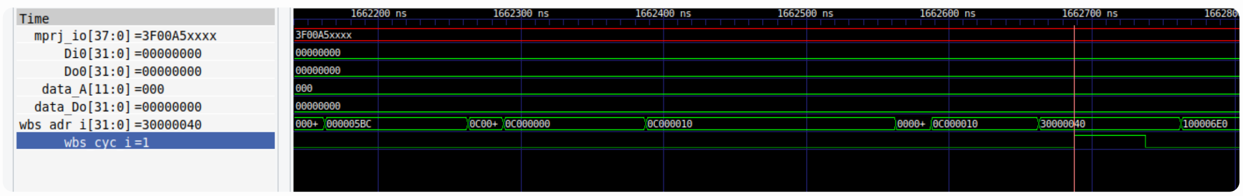
The interface protocol between firmware and user project are wishbone, and the interface protocol between testbench, firmware and user project are through mprj_pin

Waveform and analysis of the hardware/software behavior.

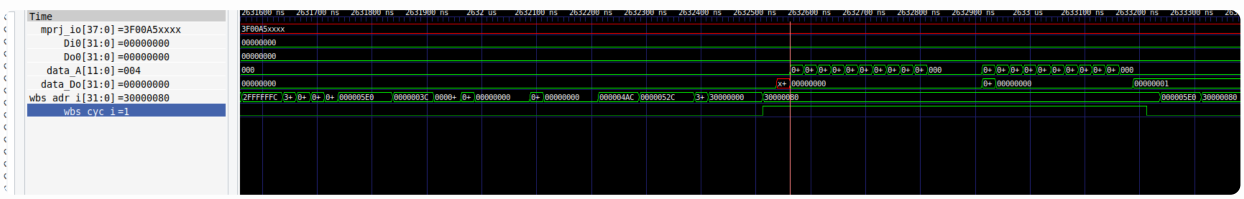
The fir function had ran three time to confirm the Y value is consistance.



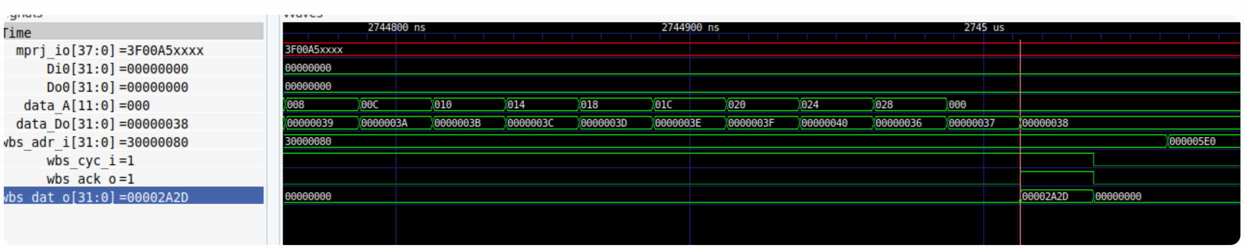
Load TAP



Load X_data



Read Y_OUT

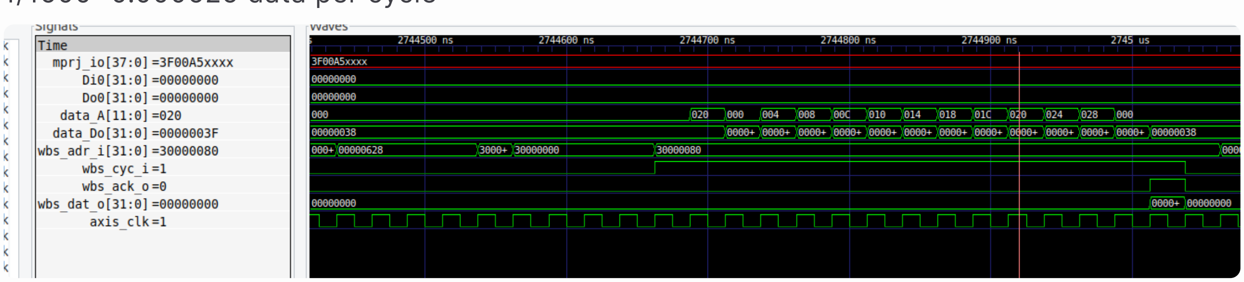


What is the FIR engine theoretical throughput, i.e. data rate? Actually

measured throughput?

theoretical throughput for Y_out is 25 cycles with data length of 64,

1/1600=0.000625 data per cycle



However, the actual measured delay is 1/10365 =0.00009647853 data per cycle

```
FIR Test 1 Started
FIR Test 1 Passed, Latency is       10365
```

The latency is calculated by testbench using counter From start flag to End Flag

```
        // repeat m times
        for (int m = 0; m < 3; m++){
                reg_mprj_datal = 0x00BA0000;//counter_start flag
                reg_mprj_datal = 0x00A50000;
                reg_data_length=data_length;
                reg_fir_verilog = 0x01;

                for (int i = 0; i < data_length; i++)
                        reg_fir_x = *(X + i);
                if(m==3){
                        int Y = reg_fir_y;
                        if (Y == 0x2A2D)
                        reg_mprj_datal =  (Y << 24) |0x005A0000;}
                else {
                        reg_mprj_datal = 0x00AB0000;}//counter_end flag
        }

        // Y data
```

Synthesis

Utilization Report

1. Slice Logic

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs	5649	0	0	53200	10.62
LUT as Logic	5407	0	0	53200	10.16
LUT as Memory	242	0	0	17400	1.39
LUT as Distributed RAM	82	0			
LUT as Shift Register	160	0			
Slice Registers	6191	0	0	106400	5.82
Register as Flip Flop	6191	0	0	106400	5.82
Register as Latch	0	0	0	106400	0.00
F7 Muxes	168	0	0	26600	0.63
F8 Muxes	47	0	0	13300	0.35

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	_	-	-
0	_	-	Set
0	_	-	Reset
0	_	Set	-
0	_	Reset	-
0	Yes	-	-
282	Yes	-	Set
943	Yes	-	Reset
116	Yes	Set	-
4850	Yes	Reset	-

2. Slice Logic Distribution

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice	2400	0	0	13300	18.05
SLICEL	1663	0			
SLICEM	737	0			
LUT as Logic	5407	0	0	53200	10.16
using O5 output only	0				
using O6 output only	4463				
using O5 and O6	944				
LUT as Memory	242	0	0	17400	1.39
LUT as Distributed RAM	82	0			
using O5 output only	0				
using O6 output only	2				
using O5 and O6	80				
LUT as Shift Register	160	0			
using O5 output only	41				
using O6 output only	81				
using O5 and O6	38				
Slice Registers	6191	0	0	106400	5.82
Register driven from within the Slice	2960				
Register driven from outside the Slice	3231				
LUT in front of the register is unused	2003				
LUT in front of the register is used	1228				
Unique Control Sets	320		0	13300	2.41

* * Note: Available Control Sets calculated as Slice * 1, Review the Control Sets Report for more information regarding control sets.

3. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	10	0	0	140	7.14
RAMB36/FIFO*	7	0	0	140	5.00
RAMB36E1 only	7				
RAMB18	6	0	0	280	2.14
RAMB18E1 only	6				

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, e a RAMB18E1

4. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	3	0	0	220	1.36
DSP48E1 only	3				

Timing Report

Summary

```
------------------------------------------------------------------------------------------------
| Design Timing Summary
| ---------------------
------------------------------------------------------------------------------------------------

    WNS(ns)      TNS(ns)  TNS Failing Endpoints  TNS Total Endpoints      WHS(ns)      THS(ns)  THS Failing Endpoints  THS Total Endpoints     WPWS(ns)     TPWS(ns)  TPWS Failing Endpoints
 TPWS Total Endpoints
    -------      -------  ---------------------  -------------------      -------      -------  ---------------------  -------------------     --------     --------  ----------------------
 --------------------
     41.118        0.000                      0                13527        0.030        0.000                      0                13527       48.750        0.000                       0
               5366


All user specified timing constraints are met.
```

Max delay

```
Max Delay Paths
--------------------------------------------------------------------------------------
Slack (MET) :              41.118ns  (required time - arrival time)
  Source:                  design_1_i/processing_system7_0/inst/PS7_i/FCLKCLK[0]
                             (clock source 'clk_fpga_0'  {rise@0.000ns fall@50.000ns period=100.000ns})
  Destination:             design_1_i/caravel_0/inst/housekeeping/wb_dat_o_reg[31]/D
                             (rising edge-triggered cell FDSE clocked by clk_fpga_0  {rise@0.000ns fall@50.000ns period=100.000ns})
  Path Group:              clk_fpga_0
  Path Type:               Setup (Max at Slow Process Corner)
  Requirement:             50.000ns  (clk_fpga_0 rise@100.000ns - clk_fpga_0 fall@50.000ns)
  Data Path Delay:         10.119ns  (logic 1.265ns (12.501%)  route 8.854ns (87.499%))
  Logic Levels:            8  (BUFG=1 LUT2=1 LUT3=1 LUT6=4 MUXF7=1)
  Clock Path Skew:         2.662ns (DCD - SCD + CPR)
    Destination Clock Delay (DCD):    2.662ns = ( 102.662 - 100.000 )
    Source Clock Delay      (SCD):    0.000ns = ( 50.000 - 50.000 )
    Clock Pessimism Removal (CPR):    0.000ns
  Clock Uncertainty:       1.500ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE
    Total System Jitter     (TSJ):    0.071ns
    Total Input Jitter      (TIJ):    3.000ns
    Discrete Jitter          (DJ):    0.000ns
    Phase Error              (PE):    0.000ns

    Location             Delay type                Incr(ns)  Path(ns)    Netlist Resource(s)
  -------------------------------------------------------------------    -------------------
                         (clock clk_fpga_0 fall edge)
                                                     50.000    50.000 f
    PS7_X0Y0             PS7                          0.000    50.000 f  design_1_i/processing_system7_0/inst/PS7_i/FCLKCLK[0]
                         net (fo=1, routed)           1.193    51.193    design_1_i/processing_system7_0/inst/FCLK_CLK_unbuffered[0]
    BUFGCTRL_X0Y21       BUFG (Prop_bufg_I_O)         0.101    51.294 f  design_1_i/processing_system7_0/inst/buffer_fclk_clk_0.FCLK_CLK_0_BUFG/O
                         net (fo=5369, routed)        2.083    53.377    design_1_i/caravel_0/inst/gpio_control_in_1[7]/clock
    SLICE_X44Y90         LUT6 (Prop_lut6_I3_O)        0.124    53.501 f  design_1_i/caravel_0/inst/gpio_control_in_1[7]/mprj_o[15]_INST_0/O
                         net (fo=2, routed)           0.442    53.943    design_1_i/caravel_ps_0/inst/control_s_axi_U/mprj_out[15]
    SLICE_X44Y90         LUT3 (Prop_lut3_I0_O)        0.124    54.067 f  design_1_i/caravel_ps_0/inst/control_s_axi_U/mprj_in[15]_INST_0/O
                         net (fo=1, routed)           0.403    54.470    design_1_i/caravel_0/inst/housekeeping/hkspi/mprj_i[15]
    SLICE_X45Y92         LUT6 (Prop_lut6_I0_O)        0.124    54.594 f  design_1_i/caravel_0/inst/housekeeping/hkspi/wb_dat_o[15]_i_22/O
                         net (fo=1, routed)           1.163    55.757    design_1_i/caravel_0/inst/housekeeping/hkspi/wb_dat_o[15]_i_22_n_0
    SLICE_X61Y93         LUT6 (Prop_lut6_I0_O)        0.124    55.881 f  design_1_i/caravel_0/inst/housekeeping/hkspi/wb_dat_o[15]_i_14/O
                         net (fo=1, routed)           0.000    55.881    design_1_i/caravel_0/inst/housekeeping/hkspi/wb_dat_o[15]_i_14_n_0
    SLICE_X61Y93         MUXF7 (Prop_muxf7_I1_O)      0.217    56.098 f  design_1_i/caravel_0/inst/housekeeping/hkspi/wb_dat_o_reg[15]_i_7/O
                         net (fo=1, routed)           0.594    56.691    design_1_i/caravel_0/inst/housekeeping/hkspi/wb_dat_o_reg[15]_i_7_n_0
    SLICE_X56Y91         LUT6 (Prop_lut6_I4_O)        0.299    56.990 f  design_1_i/caravel_0/inst/housekeeping/hkspi/wb_dat_o[15]_i_3/O
                         net (fo=5, routed)           2.977    59.967    design_1_i/caravel_0/inst/housekeeping/hkspi/wbbd_busy_reg_30
    SLICE_X51Y49         LUT2 (Prop_lut2_I1_O)        0.152    60.119 f  design_1_i/caravel_0/inst/housekeeping/hkspi/wb_dat_o[31]_i_3/O
                         net (fo=1, routed)           0.000    60.119    design_1_i/caravel_0/inst/housekeeping/hkspi_n_82
    SLICE_X51Y49         FDSE                                           f  design_1_i/caravel_0/inst/housekeeping/wb_dat_o_reg[31]/D
  -------------------------------------------------------------------    -------------------

                         (clock clk_fpga_0 rise edge)
                                                    100.000   100.000 r
    PS7_X0Y0             PS7                          0.000   100.000 r  design_1_i/processing_system7_0/inst/PS7_i/FCLKCLK[0]
                         net (fo=1, routed)           1.088   101.088    design_1_i/processing_system7_0/inst/FCLK_CLK_unbuffered[0]
    BUFGCTRL_X0Y21       BUFG (Prop_bufg_I_O)         0.091   101.179 r  design_1_i/processing_system7_0/inst/buffer_fclk_clk_0.FCLK_CLK_0_BUFG/O
                         net (fo=5369, routed)        1.483   102.662    design_1_i/caravel_0/inst/housekeeping/clock
    SLICE_X51Y49         FDSE                                           r  design_1_i/caravel_0/inst/housekeeping/wb_dat_o_reg[31]/C
                         clock pessimism              0.000   102.662
                         clock uncertainty           -1.500   101.162
    SLICE_X51Y49         FDSE (Setup_fdse_C_D)        0.075   101.237    design_1_i/caravel_0/inst/housekeeping/wb_dat_o_reg[31]
  -------------------------------------------------------------------
                         required time                        101.237
                         arrival time                         -60.119
  -------------------------------------------------------------------
                         slack                                 41.118
```

Min delay

```
Min Delay Paths
--------------------------------------------------------------------------------------
Slack (MET) :             0.030ns  (arrival time - required time)
  Source:                 design_1_i/read_romcode_0/inst/grp_read_romcode_Pipeline_VITIS_LOOP_31_1_fu_84/BUS0_addr_read_reg_150_reg[26]/C
                            (rising edge-triggered cell FDRE clocked by clk_fpga_0  {rise@0.000ns fall@50.000ns period=100.000ns})
  Destination:            design_1_i/blk_mem_gen_0/U0/inst_blk_mem_gen/gnbram.gnative_mem_map_bmg.native_mem_map_blk_mem_gen/valid.cstr/ramloop[1].ram.r/prim_noinit.ram/DEVICE_7SERIES.WITH_
BMM_INFO.TRUE_DP.SIMPLE_PRIM36.TDP_SP36_NO_ECC_ATTR.ram/DIBDI[10]
                            (rising edge-triggered cell RAMB36E1 clocked by clk_fpga_0  {rise@0.000ns fall@50.000ns period=100.000ns})
  Path Group:             clk_fpga_0
  Path Type:              Hold (Min at Fast Process Corner)
  Requirement:            0.000ns  (clk_fpga_0 rise@0.000ns - clk_fpga_0 rise@0.000ns)
  Data Path Delay:        0.382ns  (logic 0.164ns (42.952%)  route 0.218ns (57.048%))
  Logic Levels:           0
  Clock Path Skew:        0.055ns (DCD - SCD - CPR)
    Destination Clock Delay (DCD):    1.268ns
    Source Clock Delay      (SCD):    0.930ns
    Clock Pessimism Removal (CPR):    0.283ns

    Location             Delay type                Incr(ns)  Path(ns)    Netlist Resource(s)
  -------------------------------------------------------------------    -------------------
                         (clock clk_fpga_0 rise edge)
                                                      0.000     0.000 r
    PS7_X0Y0             PS7                          0.000     0.000 r  design_1_i/processing_system7_0/inst/PS7_i/FCLKCLK[0]
                         net (fo=1, routed)           0.310     0.310    design_1_i/processing_system7_0/inst/FCLK_CLK_unbuffered[0]
    BUFGCTRL_X0Y21       BUFG (Prop_bufg_I_O)         0.026     0.336 r  design_1_i/processing_system7_0/inst/buffer_fclk_clk_0.FCLK_CLK_0_BUFG/O
                         net (fo=5369, routed)        0.594     0.929    design_1_i/read_romcode_0/inst/grp_read_romcode_Pipeline_VITIS_LOOP_31_1_fu_84/ap_clk
    SLICE_X8Y38          FDRE                                         r  design_1_i/read_romcode_0/inst/grp_read_romcode_Pipeline_VITIS_LOOP_31_1_fu_84/BUS0_addr_read_reg_150_reg[26]/C
  -------------------------------------------------------------------    -------------------
    SLICE_X8Y38          FDRE (Prop_fdre_C_Q)         0.164     1.094 r  design_1_i/read_romcode_0/inst/grp_read_romcode_Pipeline_VITIS_LOOP_31_1_fu_84/BUS0_addr_read_reg_150_reg[26]/Q
                         net (fo=1, routed)           0.218     1.311    design_1_i/blk_mem_gen_0/U0/inst_blk_mem_gen/gnbram.gnative_mem_map_bmg.native_mem_map_blk_mem_gen/valid.cstr/ramloo
p[1].ram.r/prim_noinit.ram/dinb[10]
    RAMB36_X0Y7          RAMB36E1                                     r  design_1_i/blk_mem_gen_0/U0/inst_blk_mem_gen/gnbram.gnative_mem_map_bmg.native_mem_map_blk_mem_gen/valid.cstr/ramloo
p[1].ram.r/prim_noinit.ram/DEVICE_7SERIES.WITH_BMM_INFO.TRUE_DP.SIMPLE_PRIM36.TDP_SP36_NO_ECC_ATTR.ram/DIBDI[10]
  -------------------------------------------------------------------    -------------------

                         (clock clk_fpga_0 rise edge)
                                                      0.000     0.000 r
    PS7_X0Y0             PS7                          0.000     0.000 r  design_1_i/processing_system7_0/inst/PS7_i/FCLKCLK[0]
                         net (fo=1, routed)           0.337     0.337    design_1_i/processing_system7_0/inst/FCLK_CLK_unbuffered[0]
    BUFGCTRL_X0Y21       BUFG (Prop_bufg_I_O)         0.029     0.366 r  design_1_i/processing_system7_0/inst/buffer_fclk_clk_0.FCLK_CLK_0_BUFG/O
                         net (fo=5369, routed)        0.902     1.268    design_1_i/blk_mem_gen_0/U0/inst_blk_mem_gen/gnbram.gnative_mem_map_bmg.native_mem_map_blk_mem_gen/valid.cstr/ramloo
p[1].ram.r/prim_noinit.ram/clkb
    RAMB36_X0Y7          RAMB36E1                                     r  design_1_i/blk_mem_gen_0/U0/inst_blk_mem_gen/gnbram.gnative_mem_map_bmg.native_mem_map_blk_mem_gen/valid.cstr/ramloo
p[1].ram.r/prim_noinit.ram/DEVICE_7SERIES.WITH_BMM_INFO.TRUE_DP.SIMPLE_PRIM36.TDP_SP36_NO_ECC_ATTR.ram/CLKBWRCLK
                         clock pessimism             -0.283     0.985
    RAMB36_X0Y7          RAMB36E1 (Hold_ramb36e1_CLKBWRCLK_DIBDI[10])
                                                      0.296     1.281    design_1_i/blk_mem_gen_0/U0/inst_blk_mem_gen/gnbram.gnative_mem_map_bmg.native_mem_map_blk_mem_gen/valid.cstr/ramloo
p[1].ram.r/prim_noinit.ram/DEVICE_7SERIES.WITH_BMM_INFO.TRUE_DP.SIMPLE_PRIM36.TDP_SP36_NO_ECC_ATTR.ram
  -------------------------------------------------------------------
                         required time                         -1.281
                         arrival time                           1.311
  -------------------------------------------------------------------
                         slack                                  0.030
```